

QTM 220 Midterm Solution: Fall 2023 (Sept. 21)

Problem 1

Suppose we store univariate sample data in a vector y of length n and we run the following code:

```
B=10000
y.bar = rep(NA,B)
for(b in 1:B){y.bar[b]= mean(sample(y,size=n,replace=T))}
```

- a) What does the following code produce?

```
mean(y)
```

sample mean

- b) What does the following code produce?

```
sd(y)
```

sample standard deviation

- c) What does the following code produce?

```
sd(y.bar)
```

Estimated standard error

- d) What does the following code produce?

```
quantile(y.bar,probs=c(.05,.95))
```

90% CI

- e) What does the following code produce?

```
mean(y) - 1.96* sd(y)/sqrt(n)
```

```
mean(y) + 1.96* sd(y)/sqrt(n)
```

95% CI

- f) How could you re-write the code in part e) using sd(y.bar)?

```
mean(y) - 1.96* sd(y.bar)
```

```
mean(y) + 1.96* sd(y.bar)
```

Problem 2

Suppose we store bivariate sample data in a data frame `dfsam` with the first column `dfsam[,1]` the outcome variable y and the second column `dfsam[,2]` the binary covariate x . The number of rows is n .

- a) What does the following code produce?

```
mean(y[x==0])
```

sample mean for the $x = 0$ observations

- b) Explain why the following code will produce the same thing.

```
lm(y~x,data=dfsam)$coef[1]
```

with a binary covariate the linear regression intercept is the mean for the $x = 0$ group

- c) What does the following code produce?

```
mean(y[x==1]) - mean(y[x==0])
```

difference in sample means

- d) Explain why the following code will produce the same thing.

```
lm(y~x,data=dfsam)$coef[2]
```

with a binary covariate the linear regression slope is a difference in means

Problem 3

Suppose you have a sample $Y_1, Y_2, \dots, Y_n$ from a population of unknown distribution, with mean μ and standard deviation σ (and therefore variance σ^2). While you do not know the population distribution, you do know that each Y_i is independent and identically distributed (i.i.d.).

Consider the following estimator for μ : $\tilde{Y} = \frac{1}{n-2} \sum_{i=1}^n Y_i$.

a) What is $E[\tilde{Y}]$?

$$\begin{aligned} E[\tilde{Y}] &= E\left[\frac{1}{n-2} \sum_{i=1}^n Y_i\right] \\ &= \frac{1}{n-2} E\left[\sum_{i=1}^n Y_i\right] \\ &= \frac{1}{n-2} \left(\sum_{i=1}^n E[Y_i]\right) \\ &= \frac{1}{n-2} \left(\sum_{i=1}^n \mu\right) \\ &= \frac{1}{n-2} n\mu \\ &= \frac{n}{n-2} \mu \end{aligned}$$

b) Is $\tilde{Y}$ biased or unbiased for μ ?

biased

c) Showing your work, calculate $Var[\tilde{Y}]$.

$$\begin{aligned} V[\tilde{Y}] &= V\left[\frac{1}{n-2} \sum_{i=1}^n Y_i\right] \\ &= \frac{1}{(n-2)^2} V\left[\sum_{i=1}^n Y_i\right] \\ &= \frac{1}{(n-2)^2} \left(\sum_{i=1}^n V[Y_i]\right) \\ &= \frac{1}{(n-2)^2} \left(\sum_{i=1}^n \sigma^2\right) \\ &= \frac{1}{(n-2)^2} n\sigma^2 \\ &= \frac{n\sigma^2}{(n-2)^2} \end{aligned}$$

Problem 4

Suppose we are studying a trait that is distributed differently in two different groups in the population. We call these groups the “low” group (characterized by Y_{low}) and the “high” group (characterized by Y_{high}). Y_{low} and Y_{high} have the following distributions:

$$Y_{low} \underset{i.i.d.}{\sim} N(\mu, \sigma^2)$$

$$Y_{high} \underset{i.i.d.}{\sim} N(2\mu, \sigma^2)$$

The parameter of interest is μ . Suppose we are able to conduct separate random samples with replacement from the groups of sizes n_{low} and n_{high} .

- a) Consider this estimator for μ : $\hat{\mu} = \sum_{j=1}^{n_{high}} a_j Y_{j,high} - \sum_{i=1}^{n_{low}} b_i Y_{i,low}$

Find the expected value of this estimator, $E[\hat{\mu}]$.

$$\begin{aligned} E[\hat{\mu}] &= E\left[\sum_{j=1}^{n_{high}} a_j Y_{j,high} - \sum_{i=1}^{n_{low}} b_i Y_{i,low}\right] \\ &= E\left[\sum_{j=1}^{n_{high}} a_j Y_{j,high}\right] - E\left[\sum_{i=1}^{n_{low}} b_i Y_{i,low}\right] \\ &= \sum_{j=1}^{n_{high}} E[a_j Y_{j,high}] - \sum_{i=1}^{n_{low}} E[b_i Y_{i,low}] \\ &= \sum_{j=1}^{n_{high}} 2\mu a_j - \sum_{i=1}^{n_{low}} \mu b_i \\ &= \mu\left(2 \sum_{j=1}^{n_{high}} a_j - \sum_{i=1}^{n_{low}} b_i\right) \end{aligned}$$

bonus) Showing your work, calculate $Var[\hat{\mu}]$.

$$\begin{aligned}
V[\hat{\mu}] &= V\left[\sum_{j=1}^{n_{high}} a_j Y_{j,high} - \sum_{i=1}^{n_{low}} b_i Y_{i,low}\right] \\
&= V\left[\sum_{j=1}^{n_{high}} a_j Y_{j,high}\right] + V\left[\sum_{i=1}^{n_{low}} b_i Y_{i,low}\right] \\
&= \sum_{j=1}^{n_{high}} V\left[a_j Y_{j,high}\right] + \sum_{i=1}^{n_{low}} V\left[b_i Y_{i,low}\right] \\
&= \sigma^2 \sum_{j=1}^{n_{high}} a_j^2 + \sigma^2 \sum_{i=1}^{n_{low}} b_i^2 \\
&= \sigma^2 \left(\sum_{j=1}^{n_{high}} a_j^2 + \sum_{i=1}^{n_{low}} b_i^2 \right)
\end{aligned}$$